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C:\Users\ABHIRAM>vagrant up
Bringing machine 'default' up with 'virtualbox' provider...
==> default: Cloning VM...
==> default: Matching MAC address for NAT networking...
==> default: Setting the name of the VM: Win32.Cyberresearch.AV
==> default: Fixed port collision for 5985 => 55985. Now on port 2200.
==> default: Clearing any previously set network interfaces...
==> default: Preparing network interfaces based on configuration...
    default: Adapter 1: nat
==> default: Forwarding ports...
    default: 5985 (guest) => 2200 (host) (adapter 1)
    default: 5986 (guest) => 55986 (host) (adapter 1)
    default: 22 (guest) => 2222 (host) (adapter 1)
==> default: Running 'pre-boot' VM customizations...
==> default: Booting VM...
==> default: Waiting for machine to boot. This may take a few minutes...
    default: WinRM address: 127.0.0.1:2200
    default: WinRM username: malboxes
    default: WinRM execution_time_limit: PT2H
    default: WinRM transport: negotiate
==> default: Machine booted and ready!
==> default: Checking for guest additions in VM...
==> default: Mounting shared folders...
    default: /Users/malboxes/Desktop/host => C:/Users/ABHIRAM

C:\Users\ABHIRAM>

```

Figure 5: Command for creating a Vagrant file

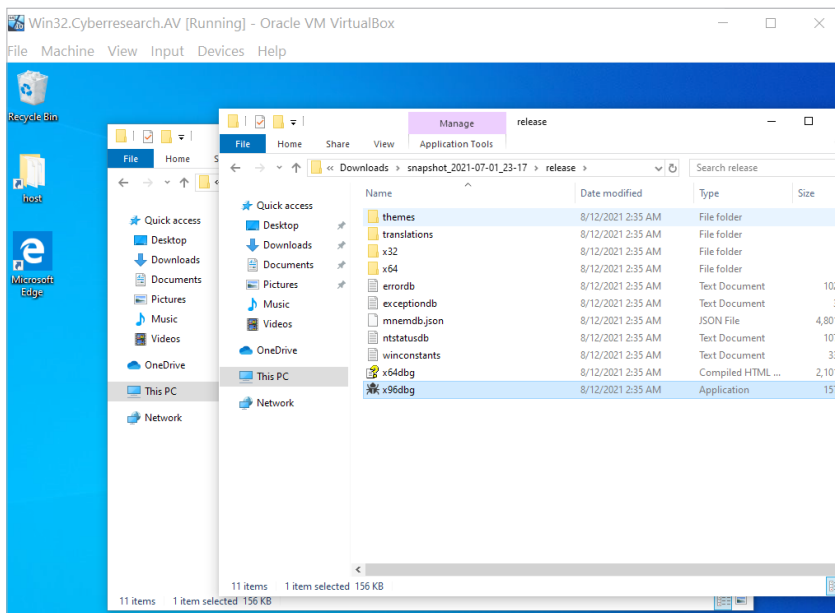


Figure 6: Sample malware for testing

- Vagrant
- VirtualBox or a vSphere/ESXi server

Minimum specs for the build machine

- At least 5GB of RAM
- VT-X extensions strongly recommended

Install VirtualBox: VirtualBox is a powerful x86 and AMD64/Intel64 virtualisation product for enterprise and home use. It is used here in order to provide a hypervisor for the Malboxes virtual machine to run on. Not only is VirtualBox an extremely rich-featured, high performance

product for enterprise customers, it is also the only solution that is freely available as open source software under the terms of the GNU General Public License (GPL) version 2.

Install Git and Python3: Git is a free and open source distributed version control system designed to handle small and very large projects with speed and efficiency. It is used to clone the Malboxes project from the GitHub repository. It is easy to learn and has a tiny footprint with lightning fast performance that outclasses SCM tools like Subversion, CVS, and Perforce.

Python is an interpreted high-level general-purpose programming language that not only constructs but also aims to help programmers write clear, logical code for small and large-scale projects with its object-oriented approach. Pip3 is the package installer for Python that can be used to install packages from the Python package index and others. It comes with the latest version of Python. We will be using Pip to install the required dependencies such as 'setuptools' and 'git install malboxes'. Pip3 gets you a newer version than that with 'apt-get'. Furthermore, combined with 'virtualenv', it allows you to install your dependencies within a single directory.

Install Vagrant: Vagrant is a virtual machine based environment management tool. Vagrant Boxes are prebuilt basic images that can be used as a starting point. This can be used to build a Malboxes image to test malware on it. Vagrant works on Mac, Linux, Windows, and more. Remote development environments force users to give up their favourite editors and programs. Vagrant works on your local system with the tools you're already familiar with.

Install Packer 1.2.3 and add it to environment variables: Packer is a program that automates the production of machine images of any form. It encourages users to utilise a framework like Chef or Puppet to install and configure software within